Beyond Political Segregation: Divided on Every Front

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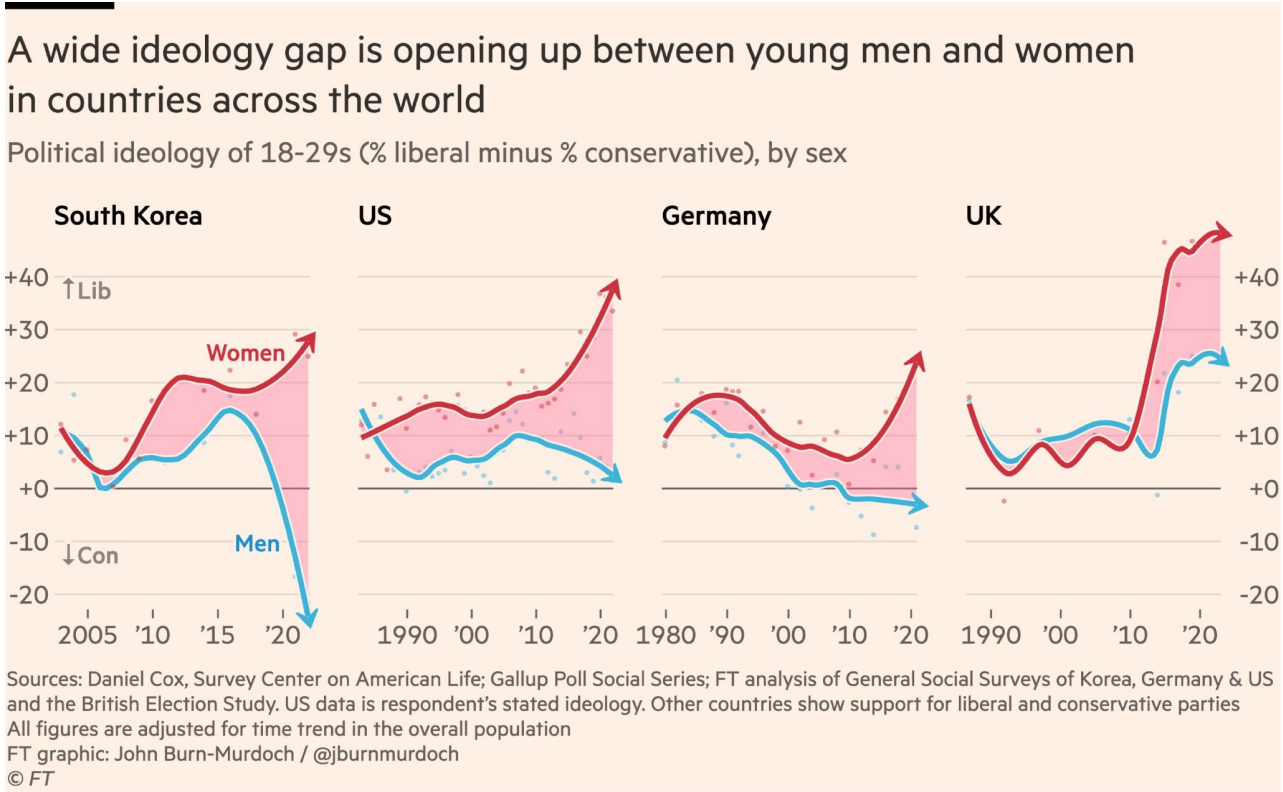
Motivation



Moon and Kim 2024

Gillion et al. 2018

Why: Background?

- 504% increase in mobile internet media consumption since 2011 (Pew 2021)

- Democratization of content creation on platforms FB/Meta, Twitter/X, and IG with low entry barrier

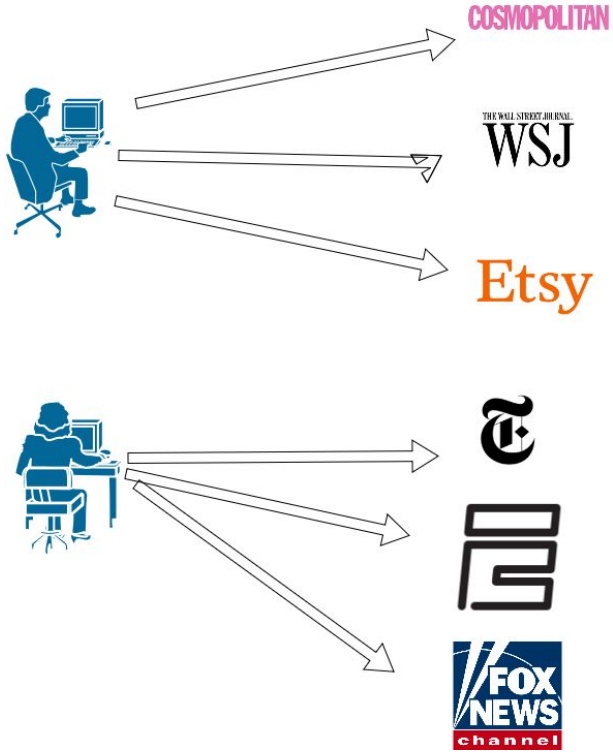
This creates individuals with information overflow and high degree of choice, and “niche audiences” (Stroud, 2011)

Before

Media creates **public sphere** (Habermas, 1989)

Discussions among diverse groups > informed and democratic decision-making. (Fishkin, 1991)

Cross-cutting interactions > reduces polarization > cohesive, democratic society. (Mutz, 2006)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THE WALL STREET JOURNAL.
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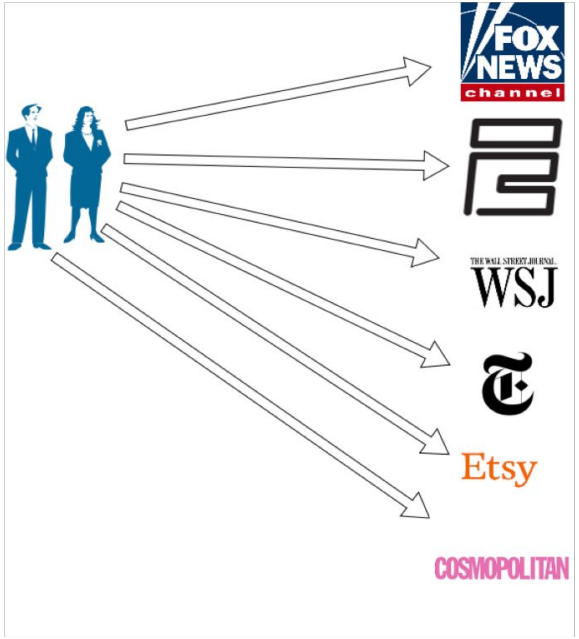
FOX NEWS channel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Debating the Impact of the Internet

- **Polarization Perspective**:
  - Internet fuels political polarization as users:
    - choose to avoid conflicting viewpoints (**Selective Exposure:** Sunstein 2002, 2017)
    - be algorithmically driven toward politically favorable content (**Filter Bubble**:Pariser 2011)
    - networked processes (**Curation Bubble**: Green et al. 2024)

  González-Bailón et al. (2023) political segregation in Facebook

- **Cross-Cutting Perspective:**
  - Politically diverse sources are extremely common (Gentzkow & Shapiro 2011; Messing & Westwood 2014; Bakshy, Messing, & Adamic 2015; Barbera 2015; Guess 2021)

Gaps in Prior Literature

- **Source Level Focus:** Domain vs URL level analysis
  - missing variations in the user information consumption (Green et al. 2024)

- **Political Segregation Focus:**
  - most people not interested in politics (Schattschneider 1960, Lupia 2016)
  - segregation across other demographic variables: age and gender (Monti et al. 2023)

Research Questions

1) **How information sharing on Twitter/X exhibit segregation at the domain and URL level?**

2) **What are the levels of this segregation based on age, gender, and partisanship?**

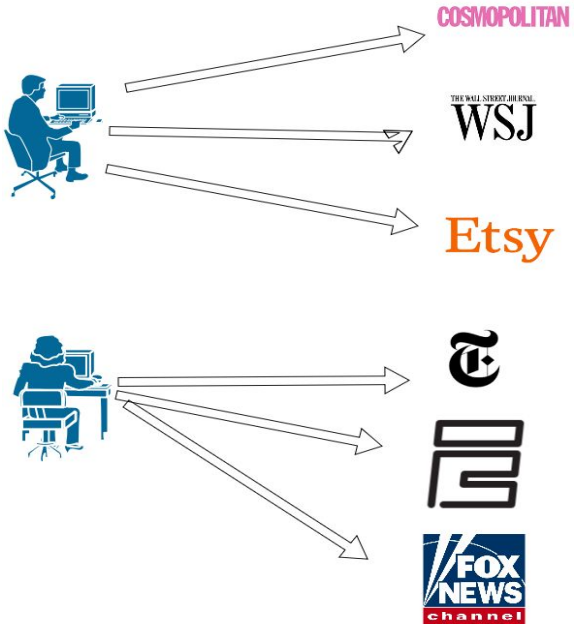
Data

- Panel of 1.6 million Twitter/X user accounts linked to US voter file (Hughes et al., 2021, Shugars et al.2021)

- 1.1M Domains and 43M URLs btw 01/2020-12/2021

username	tweet_domain	tweet_url	party	gender	age
Adam Smith	NYT	nyt.com/capitalismisbrat	Republican	male	18–29
Karl Marx	NYT	nyt.com/10reassonswhycapitalismbad	Democrat	female	+65

Data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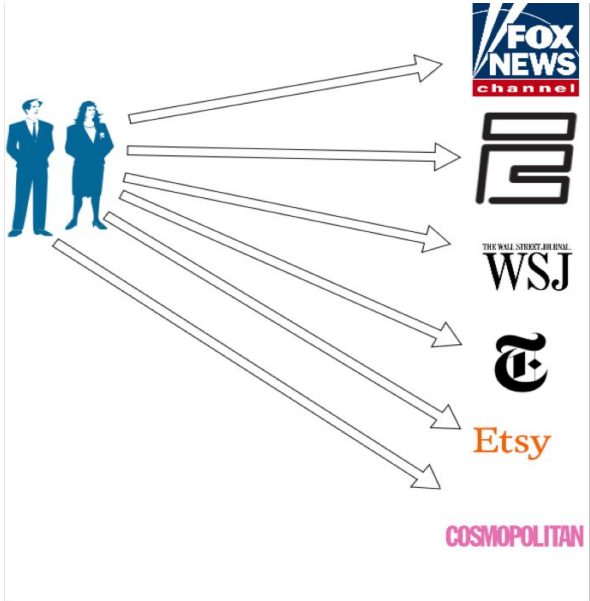
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Method: Segregation Index

$$S = \boxed{\sum_{n \in N} \left(\frac{R_n}{R} \times \frac{R_n}{v_n} \right)} - \boxed{\sum_{n \in N} \left(\frac{D_n}{D} \times \frac{R_n}{v_n} \right)}$$

R_n = Number of Conservative views in URL/domain n
D_n = Number of Liberal views in URL/domain n
v_n = Total views in URL/domain n
C = Total number of Democrat shares
L = Total number of Republican shares
N= Set of all URLs/Domains

Extent to which Dems and Reps visits the same place

0<S<1 , the higher the more segregated

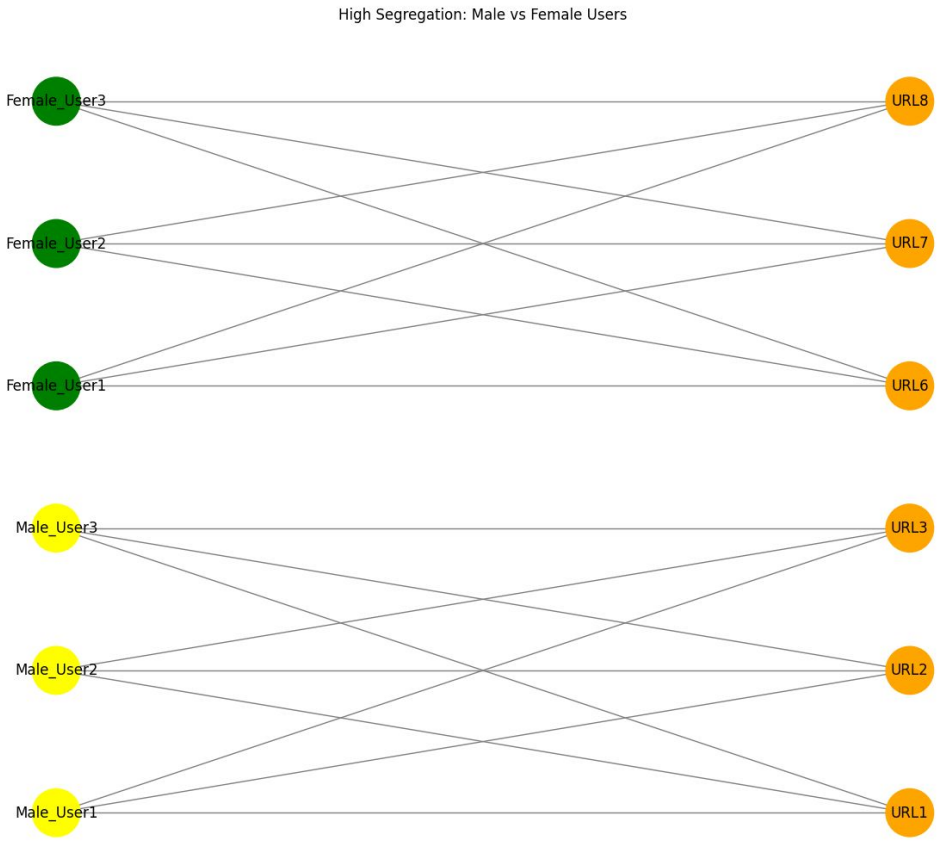
Method

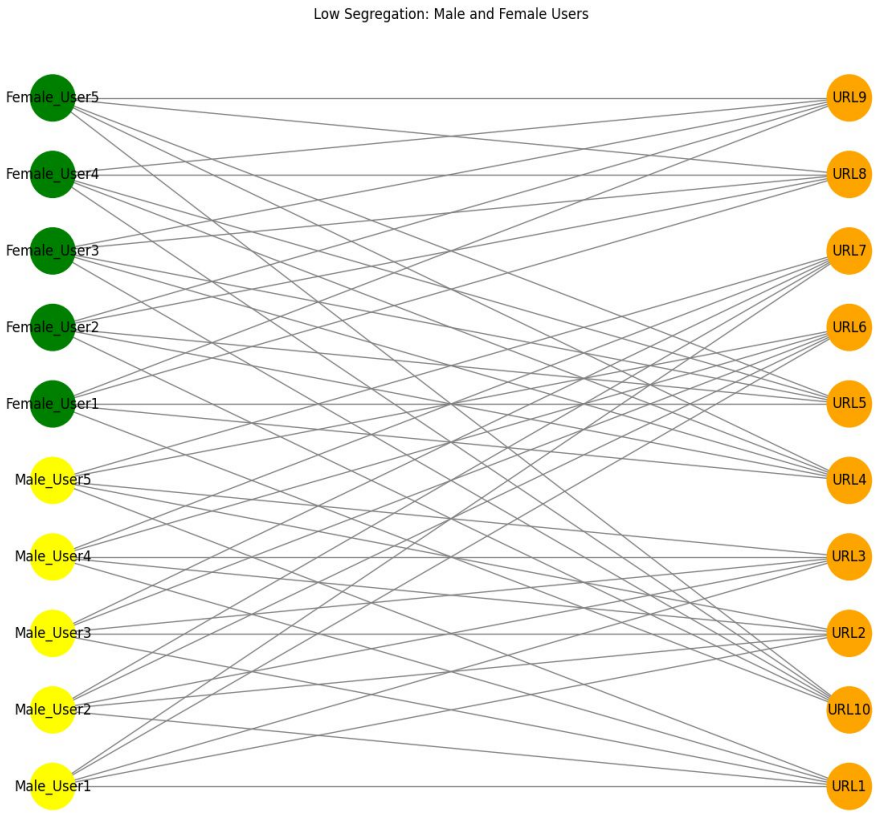
Partisanship = $$\sum_{n \in N} \left(\frac{R_n}{R} \times \frac{R_n}{v_n} \right) - \sum_{n \in N} \left(\frac{D_n}{D} \times \frac{R_n}{v_n} \right)$$

Gender = $$\sum_{n \in N} \left(\frac{M_n}{M} \times \frac{M_n}{v_n} \right) - \sum_{n \in N} \left(\frac{F_n}{F} \times \frac{M_n}{v_n} \right)$$

Age = $$\sum_{n \in N} \left(\frac{Age1_n}{Age1} \times \frac{Age1_n}{v_n} \right) - \sum_{n \in N} \left(\frac{Age2_n}{Age2} \times \frac{Age1_n}{v_n} \right)$$

Segregation Levels





Segregation Index

BROWSING < 0.1 (Gentzkow & Shapiro, 2011,Yang et al. 2020; Flaxman et al. 2016)

FB/Meta Election Study = ~0.5 for url and 0.35 for domain (Gonzalez Bailon et al. 2023)

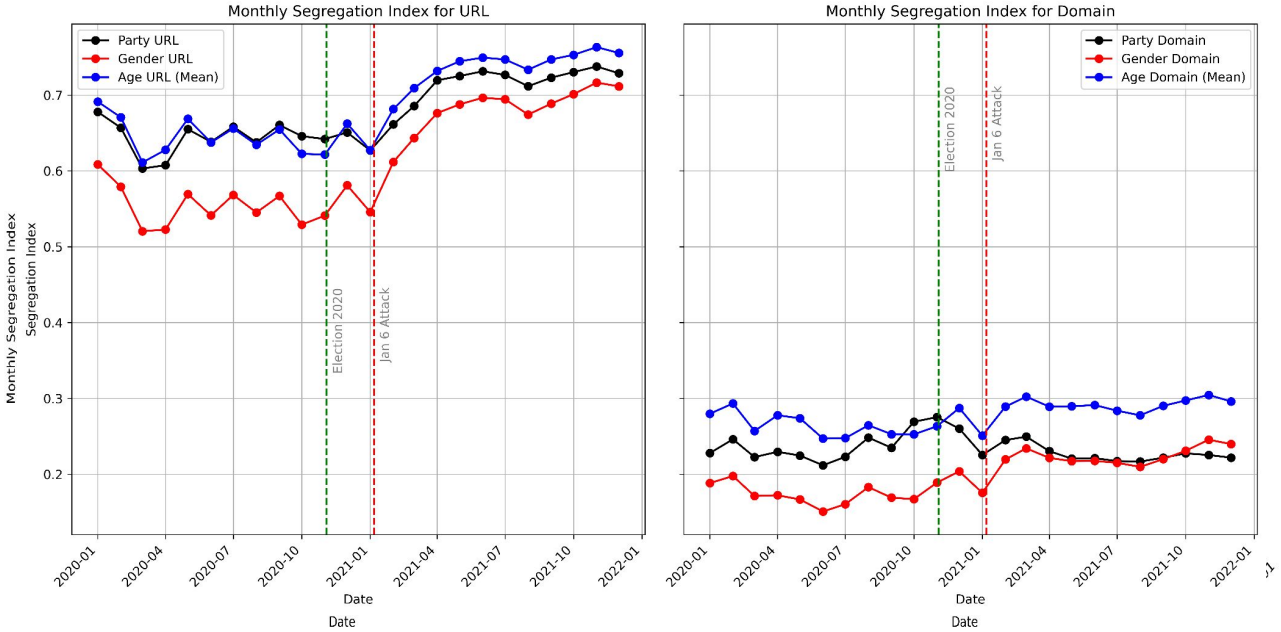
Results: Monthly Segregation

1))URL segregation > Domain

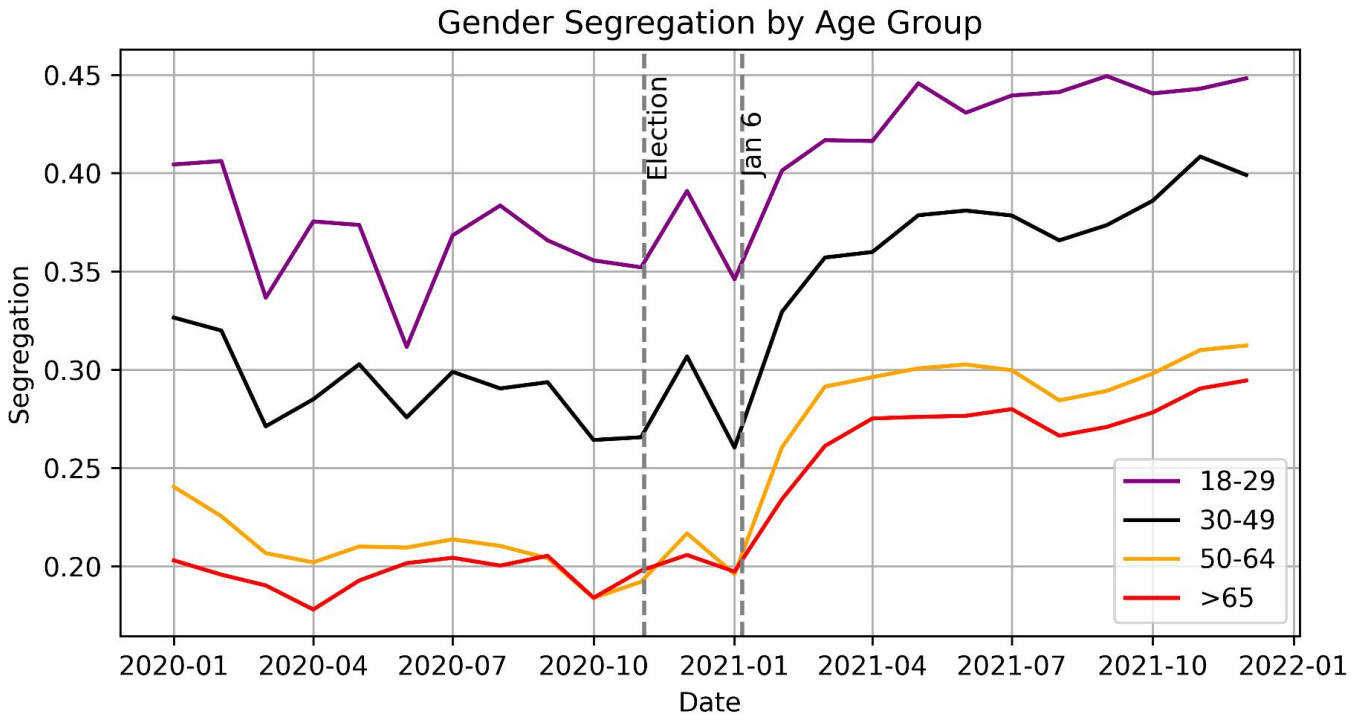
2)Age is the most segregated

3) Increase after Jan 6

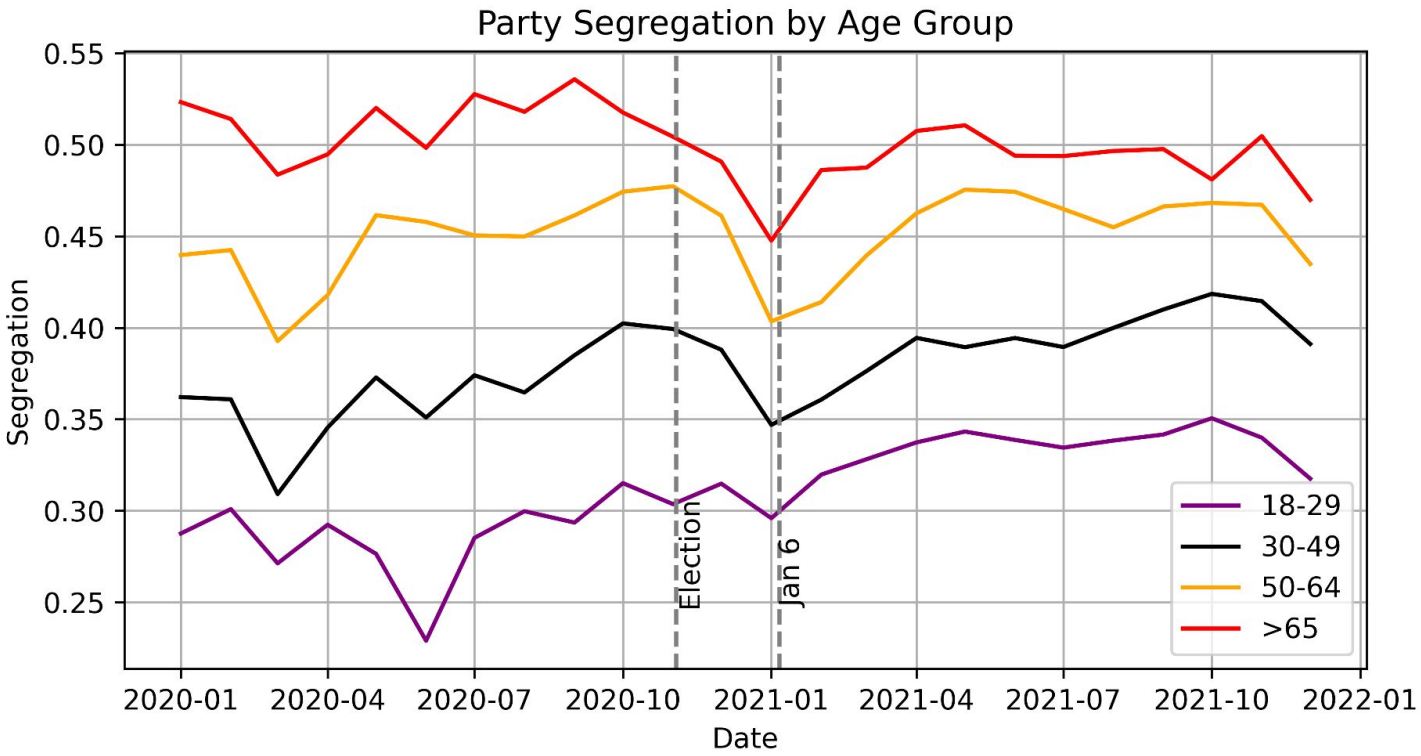
4) Twitter is more segregated than the FB (S= 0.5 González-Bailón et al. (2023))



Gender


Gender Segregation by Age Group
Segregation
0.45
0.40
0.35
0.30
0.25
0.20
Election
Jan 6
18-29
30-49
50-64
>65
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01
Date

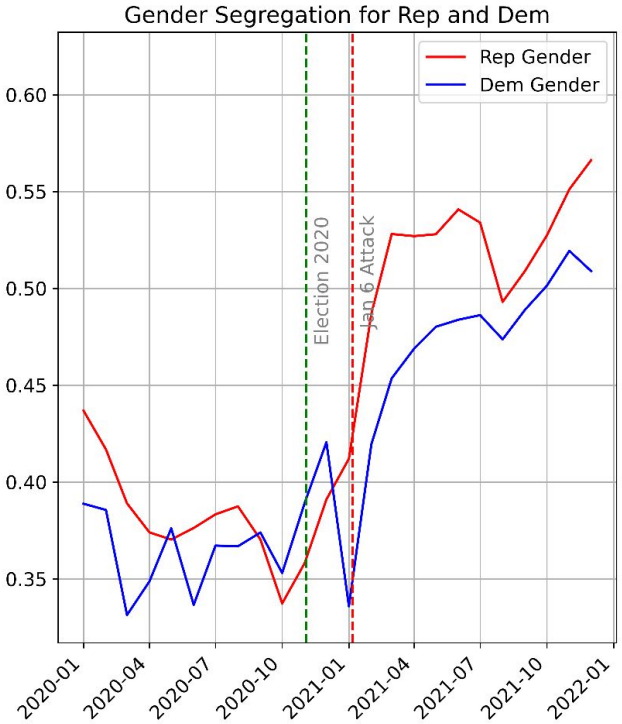
Party

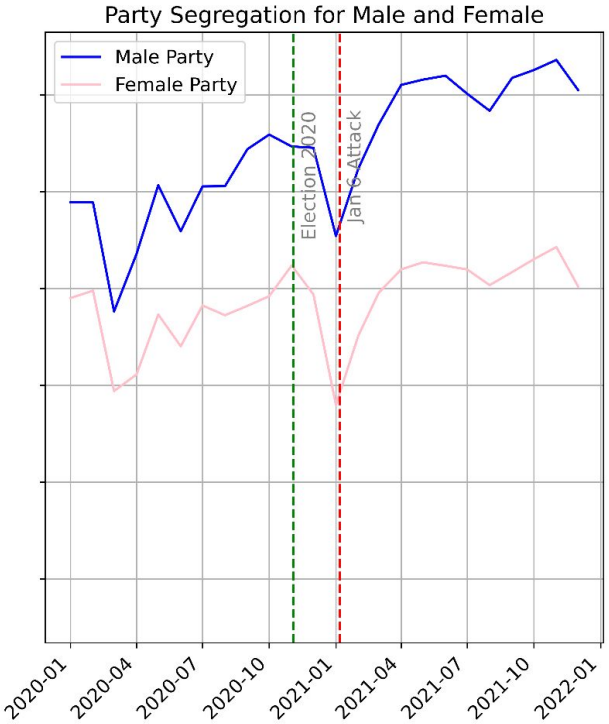

Party Segregation by Age Group
Segregation
0.55
0.50
0.45
0.40
0.35
0.30
0.25
Election
Jan 6
18-29
30-49
50-64
>65
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01
Date

Age

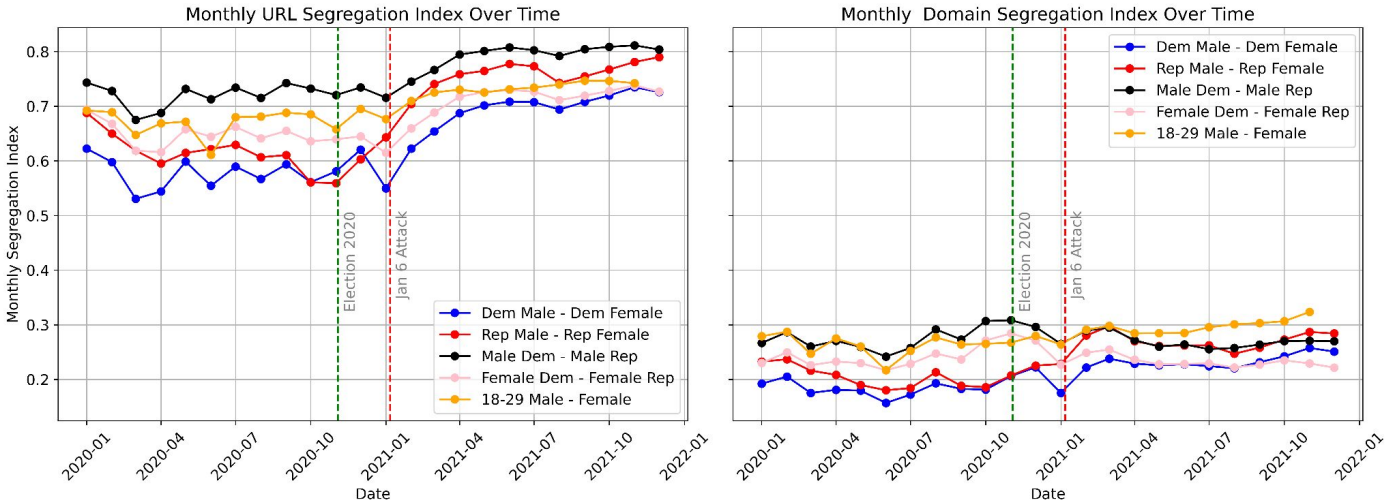
Segregation of Age 18-29
Segregation of Age >65
Segregation of Age 30-49
Segregation of Age 50-64
Gender
Partisanship
Election
Jan 6
0.6
0.5
0.4
0.3
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01

Gender


Gender Segregation for Rep and Dem
Rep Gender
Dem Gender
Election 2020
Jan 6 Attack
0.60
0.55
0.50
0.45
0.40
0.35
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01


Party Segregation for Male and Female
Male Party
Female Party
Election 2020
Jan 6 Attack
2020-01
2020-04
2020-07
2020-10
2021-01
2021-04
2021-07
2021-10
2022-01

Demographic Interactions

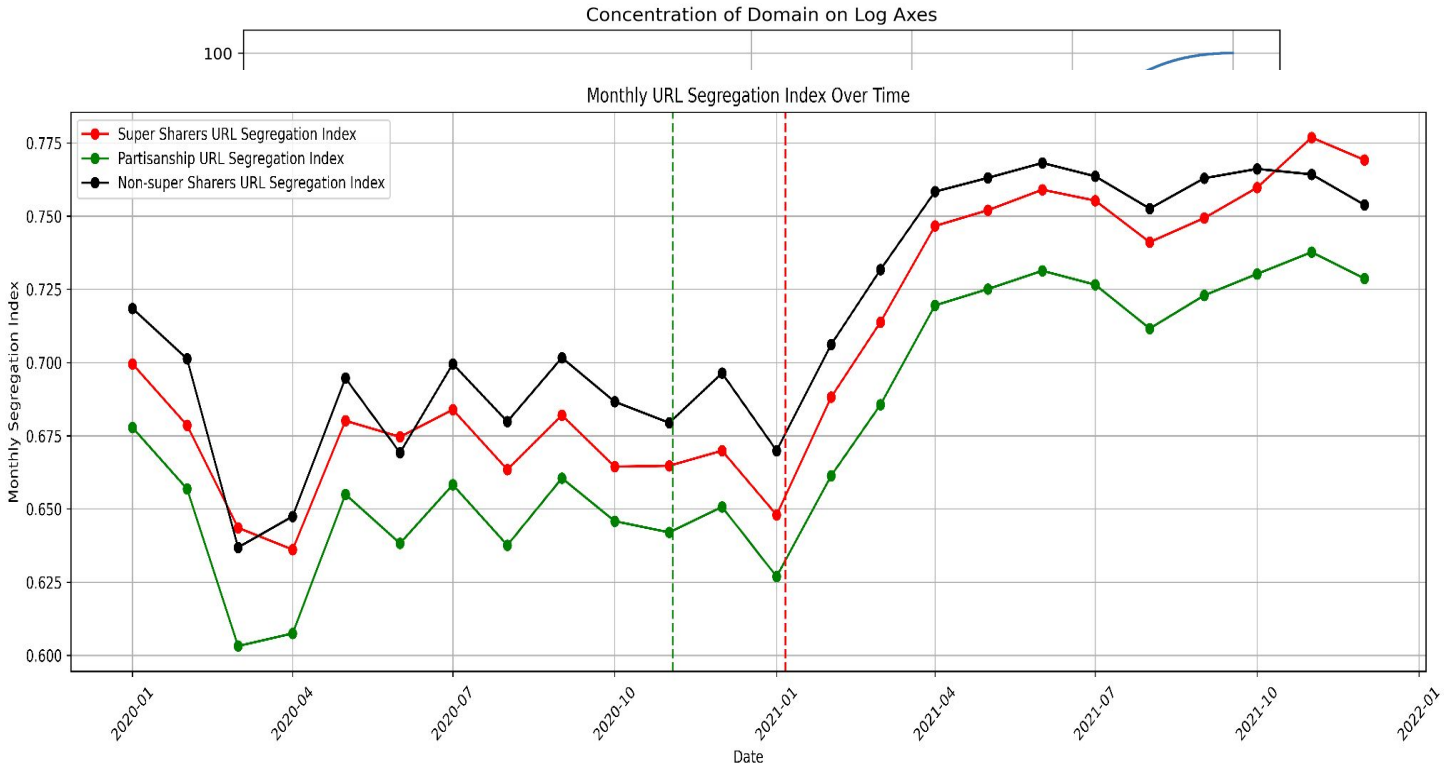


Republican Male and Democrat Male is the most segregated (black line)

S (R female and R male) increased the most

Segregation of Super sharers

Super sharers are less segregated:



Some Domains

Parsed_Domain	High_Male_Share_Male_Count	male_share_percentage
cleverprogrammer.to	12073	100.0
smpt.co	4140	100.0
PSNProfiles.com	2287	100.0
maizenbrew.com	1714	100.0
thecrochetdude.com	1612	100.0
colltales.com	1589	100.0
couchguysports.com	2036	99.95090819833090
zagsblog.com	2031	99.9507874015748
morningjournal.com	1747	99.94279176201370
magamedia.org	2783	99.89231873653990
dailybible.co	1592	99.87452948557090
canalstreetchronicles.com	2554	99.84362783424550
newsroom.co.nz	1721	99.8259860788863
endtimeheadlines.org	2130	99.76580796252930
wtov9.com	2237	99.42222222222220
thednvr.com	1568	99.303356554781
lauraloomerforcongress.com	1649	98.86091127098320
pff.com	2167	98.7243735763098
theblast.com	2571	98.58128834355830
fanbox.cc	1841	98.18666666666670
justfor.fans	6401	97.71027324072660

	Parsed_Domain	Female_Count	Female_share_percentage
2	candidceillie.com	1466	100.0
3	SouthernHerbalRemedies.com	1429	100.0
4	courtneyleedesigns.com	1157	100.0
5	littledogtips.com	1117	100.0
6	strengthandsunshine.com	957	100.0
7	empiricaldata.org	916	100.0
8	relevantradio.com	828	100.0
9	travelweekly.com	1439	99.79195561719830
10	colourpop.com	1537	99.09735654416510
11	turnto10.com	2113	98.87693027608800
12	alabamawx.com	1310	98.49624060150380
13	lbpost.com	1553	98.41571609632450
14	poshmark.com	31516	98.13482796201150
15	bethesdamagazine.com	1316	97.98957557706630
16	weverse.io	896	97.81659388646290
17	koreaboo.com	1207	97.49596122778680
18	wboy.com	929	97.4816369359916
19	liveme.com	2040	96.91211401425180
20	soompi.com	2898	96.18320610687020

Takeaways!!!

Twitter/X is highly segregated!!!

Age> Partisanship > Gender

To-Dos

Null effect vs real segregation numbers

Platform affordances: Sharing vs consumption

Questions

Thank you!!!!!

Contact: azturan.b@northeastern.edu

Grab me to talk further :)

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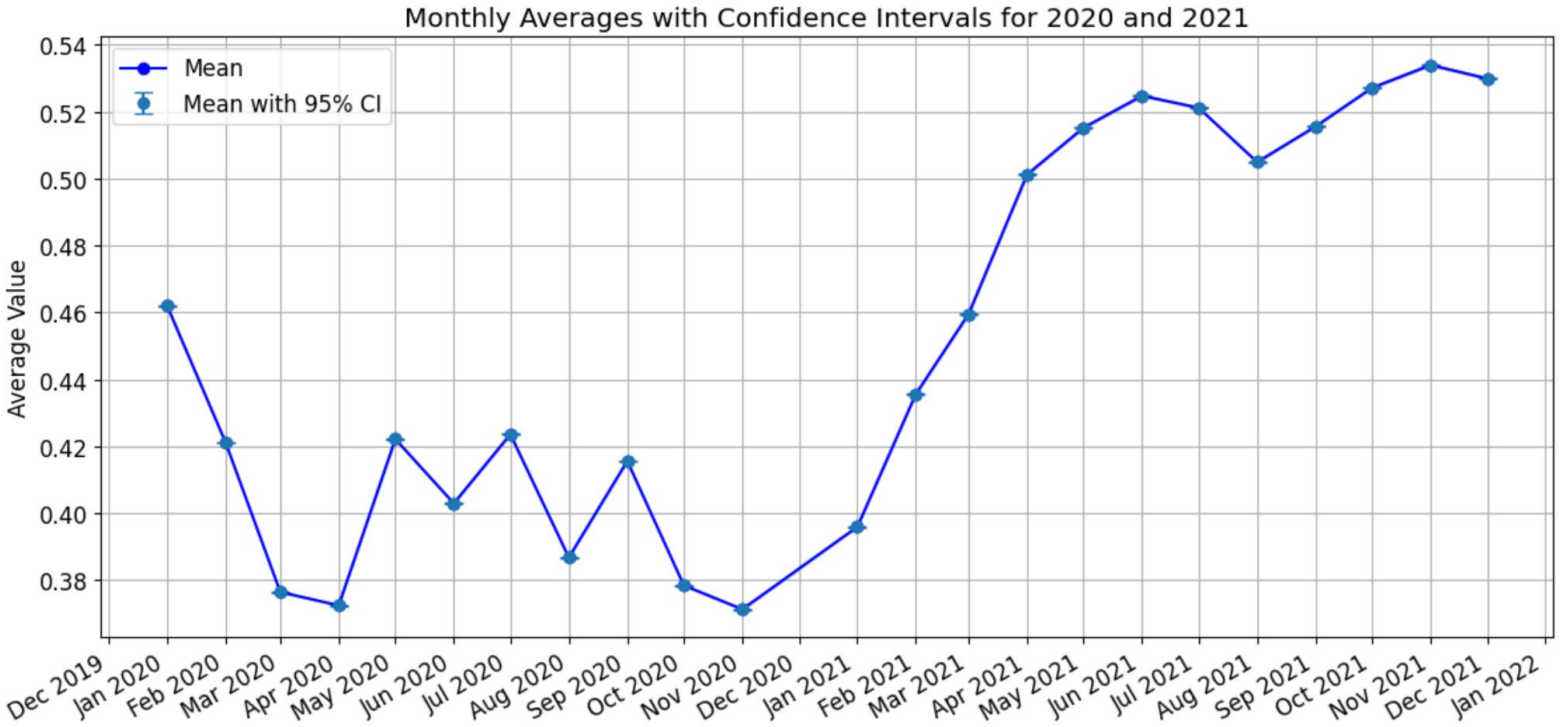
Random Permutation



Data (i): Twitter Users

10%

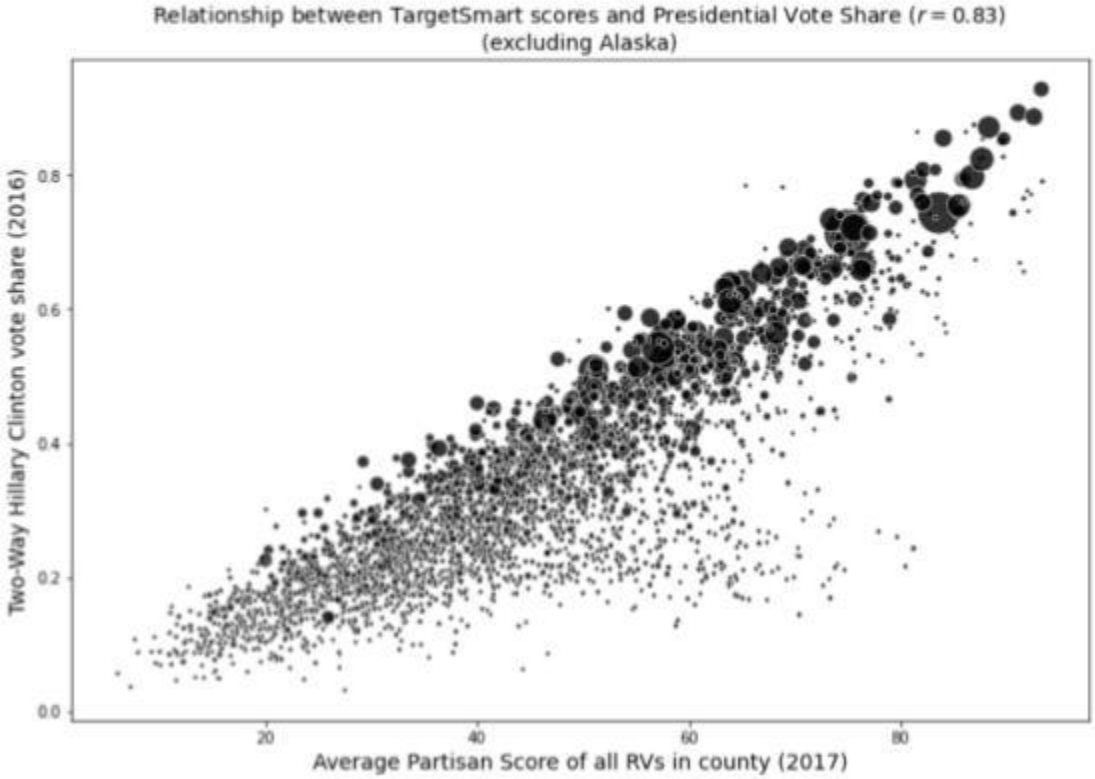
>406 million
(Decahose
1/2014-3/2017)

>322 million
(active)

Stefan McCabe
@mccabe_s

Econlib
@Econlib

290 million
(have names)

1.5 million
(linked to
voter data)

TargetSmart Partisanship score